

Optimal Radial Equalization for Modular Frames with Central Lengths

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2026-08-11

Source question and result

Let $\mathcal{A}$ be a unital C^* -algebra and give the standard left module $\mathcal{A}^d$ the inner product

$$\langle x, y \rangle = \sum_{k=1}^d x_k y_k^*.$$

Problem 2.10 of Krishna, arXiv:2207.12799, asks for the closest equal-inner-product modular frame to a given ε -nearly equal-inner-product modular frame $\{\tau_j\}_{j=1}^n$, where

$$(1 - \varepsilon) \frac{d}{n} 1 \leq p_j := \langle \tau_j, \tau_j \rangle \leq (1 + \varepsilon) \frac{d}{n} 1 \quad (1 \leq j \leq n) \quad (1)$$

and $0 < \varepsilon < 1$. The squared modular distance is

$$\text{dist}(\tau, \omega)^2 = \left\| \sum_{j=1}^n \langle \tau_j - \omega_j, \tau_j - \omega_j \rangle \right\|.$$

The theorem below completely answers the question when the squared lengths p_j are central, in particular for every commutative C^* -algebra. For arbitrary $\mathcal{A}$ it still gives an explicit equal-inner-product frame, exact distance, and a dimensionally sharp quadratic-in- ε bound.

Idea of the Proof

Normalize each vector along its own left-module ray. Functional calculus makes the resulting length exactly $\sqrt{d/n}$ and controls the new frame bounds. To prove optimality, normalize any competitor as well. The cross inner product of two unit module vectors is a contraction. When p_j is central, its square root can pass through the cross term, so the elementary inequality $q + q^* \leq 2$ becomes a positive-element lower bound for each competitor displacement. The radial choice attains every one of these lower bounds simultaneously.

Theorem 1 (radial equalization and central-length optimality). *Set $c = \sqrt{d/n}$ and define*

$$\omega_j = c p_j^{-1/2} \tau_j \quad (1 \leq j \leq n). \quad (2)$$

Then the following hold.

1. $\{\omega_j\}_{j=1}^n$ is an equal-inner-product modular frame. If $\{\tau_j\}$ has frame bounds a, b , then $\{\omega_j\}$ has frame bounds

$$\frac{a}{1+\varepsilon}, \quad \frac{b}{1-\varepsilon}.$$

2. Its distance is exactly

$$\text{dist}(\tau, \omega)^2 = \left\| \sum_{j=1}^n (p_j^{1/2} - c1)^2 \right\|, \quad (3)$$

and consequently

$$\text{dist}(\tau, \omega)^2 \leq d(1 - \sqrt{1-\varepsilon})^2 \leq d\varepsilon^2. \quad (4)$$

3. If every p_j belongs to the center $Z(\mathcal{A})$, then $\{\omega_j\}$ is globally closest among all equal-inner-product modular frames. Indeed, for every collection $\{\nu_j\}$ satisfying $\langle \nu_j, \nu_j \rangle = c^2 1$,

$$\text{dist}(\tau, \nu)^2 \geq \text{dist}(\tau, \omega)^2. \quad (5)$$

Proof. By (1), each p_j is positive and invertible. With $a_j = cp_j^{-1/2}$, the left-module convention gives

$$\langle \omega_j, \omega_j \rangle = a_j p_j a_j = c^2 1.$$

Furthermore,

$$\frac{1}{1+\varepsilon} 1 \leq a_j^2 \leq \frac{1}{1-\varepsilon} 1.$$

For every $x \in \mathcal{A}^d$,

$$\langle x, \omega_j \rangle \langle \omega_j, x \rangle = \langle x, \tau_j \rangle a_j^2 \langle \tau_j, x \rangle.$$

Sandwiching the displayed order bounds and then using the original frame inequalities proves the new frame bounds.

Since $1 - cp_j^{-1/2}$ is a self-adjoint function of p_j ,

$$\langle \tau_j - \omega_j, \tau_j - \omega_j \rangle = (1 - cp_j^{-1/2}) p_j (1 - cp_j^{-1/2}) = (p_j^{1/2} - c1)^2,$$

which proves (3). The spectrum of $p_j^{1/2}$ lies in

$$[c\sqrt{1-\varepsilon}, c\sqrt{1+\varepsilon}].$$

For $0 < \varepsilon < 1$ the larger endpoint deviation is $c(1 - \sqrt{1-\varepsilon})$. Hence each summand in (3) is at most $c^2(1 - \sqrt{1-\varepsilon})^2 1$. Summing, using $nc^2 = d$, and observing

$$1 - \sqrt{1-\varepsilon} = \frac{\varepsilon}{1 + \sqrt{1-\varepsilon}} \leq \varepsilon$$

gives (4).

It remains to prove optimality under centrality. Put

$$v_j = p_j^{-1/2} \tau_j, \quad u_j = c^{-1} \nu_j, \quad q_j = \langle v_j, u_j \rangle.$$

Then $\langle v_j, v_j \rangle = \langle u_j, u_j \rangle = 1$. The C^* -module Cauchy–Schwarz inequality gives $\|q_j\| \leq 1$, whence $q_j + q_j^* \leq 21$. Since $p_j^{1/2}$ is central, expansion in the left-module convention yields

$$\begin{aligned} \langle \tau_j - \nu_j, \tau_j - \nu_j \rangle &= p_j + c^2 1 - cp_j^{1/2} (q_j + q_j^*) \\ &\geq p_j + c^2 1 - 2cp_j^{1/2} = (p_j^{1/2} - c1)^2. \end{aligned}$$

Summing preserves order. Norm is monotone on positive elements, so (5) follows from (3). The radial collection attains equality and is a frame by the first part. $\square$

Scope and obstruction

The centrality hypothesis is automatic when $\mathcal{A}$ is commutative, but it is not cosmetic in the proof. Without it, the two cross terms are

$$cp_j^{1/2}q_j \quad \text{and} \quad cq_j^*p_j^{1/2};$$

they cannot be combined as positive multiplication of $q_j + q_j^* \leq 2$. Composing with a state does not repair this, because a general state is not tracial. Likewise the modular distance is a C^* -norm of an $\mathcal{A}$ -valued sum, not a scalar unitarily invariant Hilbert–Schmidt norm. Thus Theorem 1 is a full solution of Problem 2.10 for central squared lengths and a controlled feasible solution in general; the optimal noncentral frame remains open.

Eight focused upgrade routes were pursued: duplicate checking against the paper’s appendix, arbitrary-algebra radial normalization, exact frame and distance estimates, a commutative fiber-wise proof, the central-order upgrade, statewise lower bounds, polar-decomposition analogies, and alternating length/frame normalization. The last three all meet the same opposite-sided cross-term obstruction.

Novelty check

Exact arXiv-id, title, “closest equal inner product modular frame,” and “Modular Paulsen Problem” searches were run in the local FA/Banach indexes and primary arXiv sources through 2026-08-11. No later answer to Problem 2.10 and no existing run packet for it were found. The appendix restricted-invertibility and literally stated Johnson–Lindenstrauss questions are already handled by the separate GPT5.6 packet attached to arXiv:2208.05223 and are not duplicated here.

References

- K. Mahesh Krishna, *Modular Paulsen Problem and Modular Projection Problem*, arXiv:2207.12799.
- M. Frank and D. R. Larson, *Frames in Hilbert C^* -modules and C^* -algebras*, J. Operator Theory 48 (2002), 273–314.